

CHALLENGES OF USING WATERFALL IN ANALYTICS PROJECTS: CHANGING REQUIREMENTS, DELAYED FEEDBACK, AND DATA UNCERTAINTY

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WHY WATERFALL AND ANALYTICS CONFLICT

MOTIVATION

- Analytics projects are critical for organizational decision making
- Many analytics projects are still managed using Waterfall
- Waterfall was designed for stable and predictable environments
- Analytics work is exploratory and uncertain
- This creates a structural mismatch



RESEARCH QUESTION **AND THESIS**

**Why does Waterfall
struggle when
applied to analytics
projects?**

1

Changing requirements

2

Delayed feedback

3

Data uncertainty

BACKGROUND FROM LITERATURE

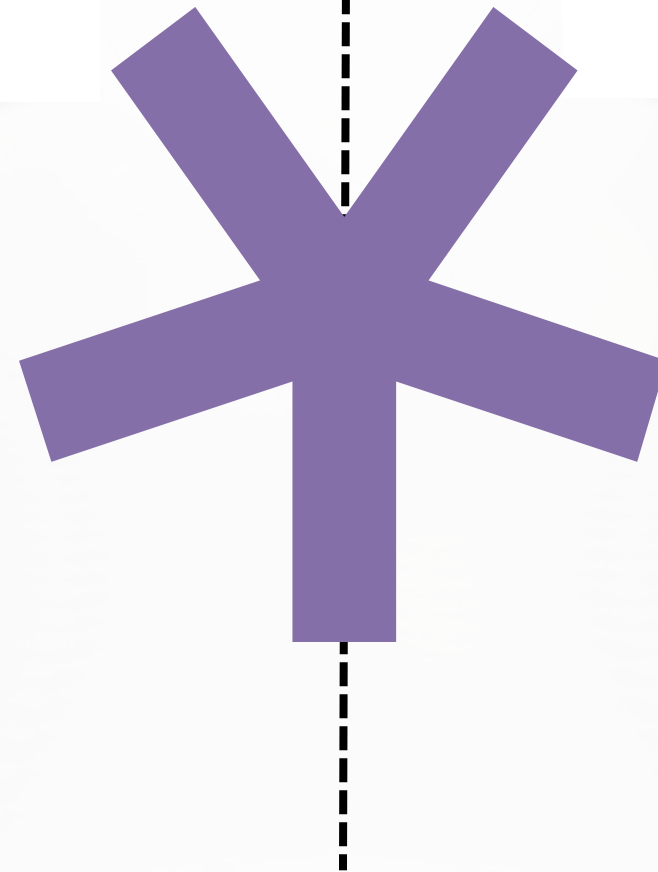
Waterfall emphasizes linear phases and early planning

Analytics literature emphasizes iteration and learning

Works well when requirements and inputs are stable

Data science frameworks assume evolving understanding

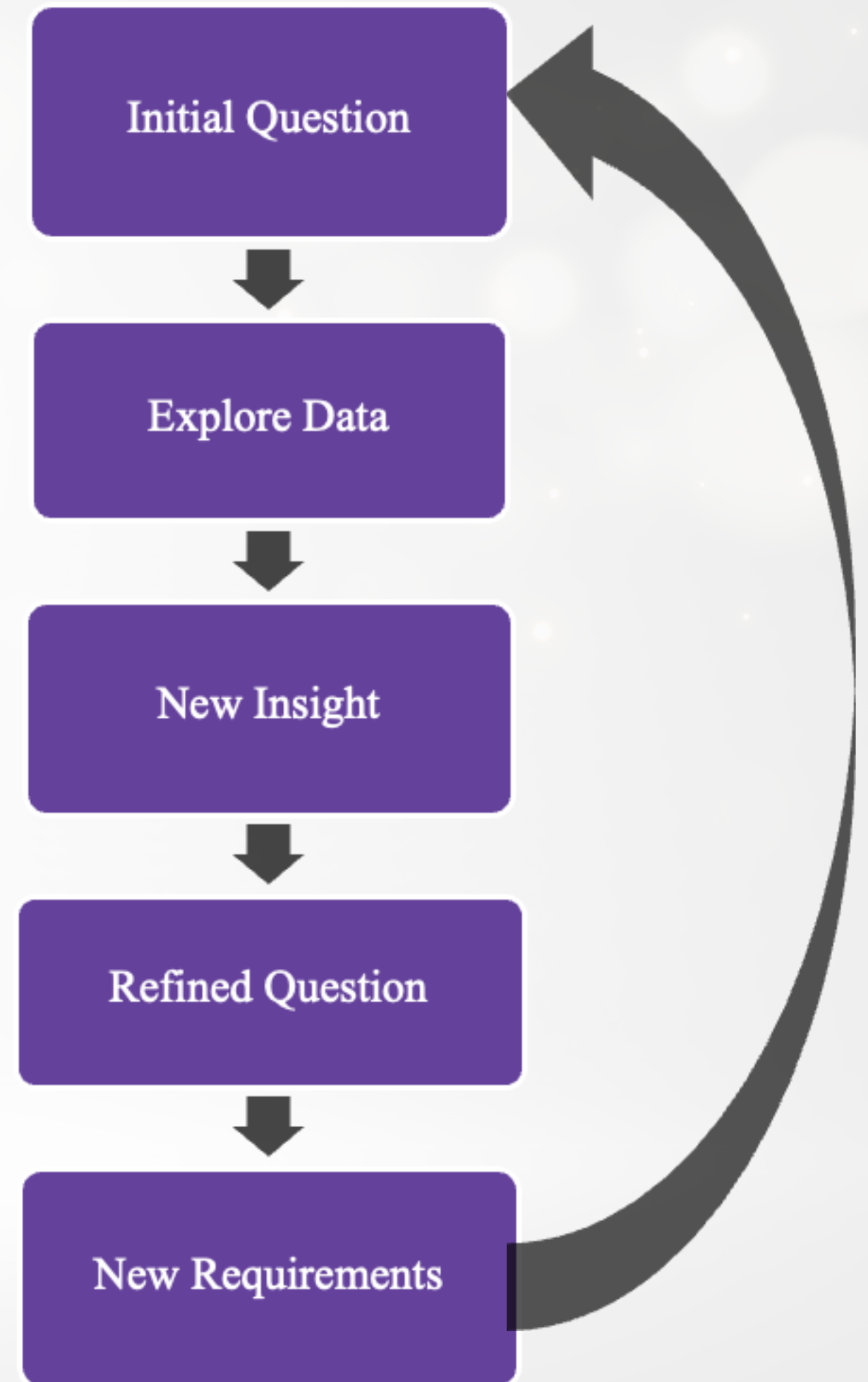
These two approaches are fundamentally different



ARGUMENT 01

CHANGING REQUIREMENTS

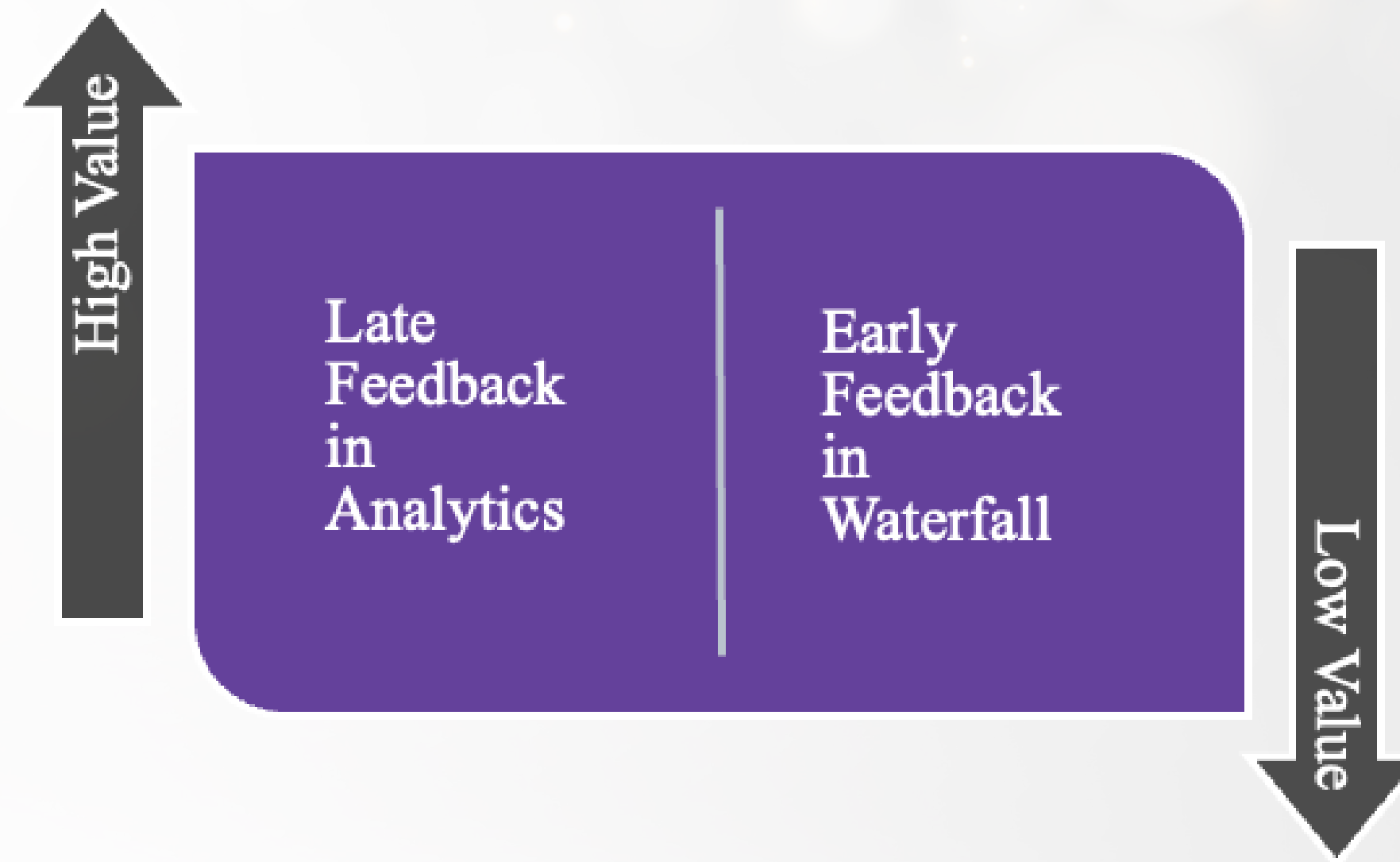
- Waterfall assumes requirements can be defined upfront
- Analytics requirements evolve through data exploration
- Stakeholders refine goals after seeing results
- New insights change project direction
- Changes cause rework and delays in Waterfall



ARGUMENT 02

DELAYED FEEDBACK

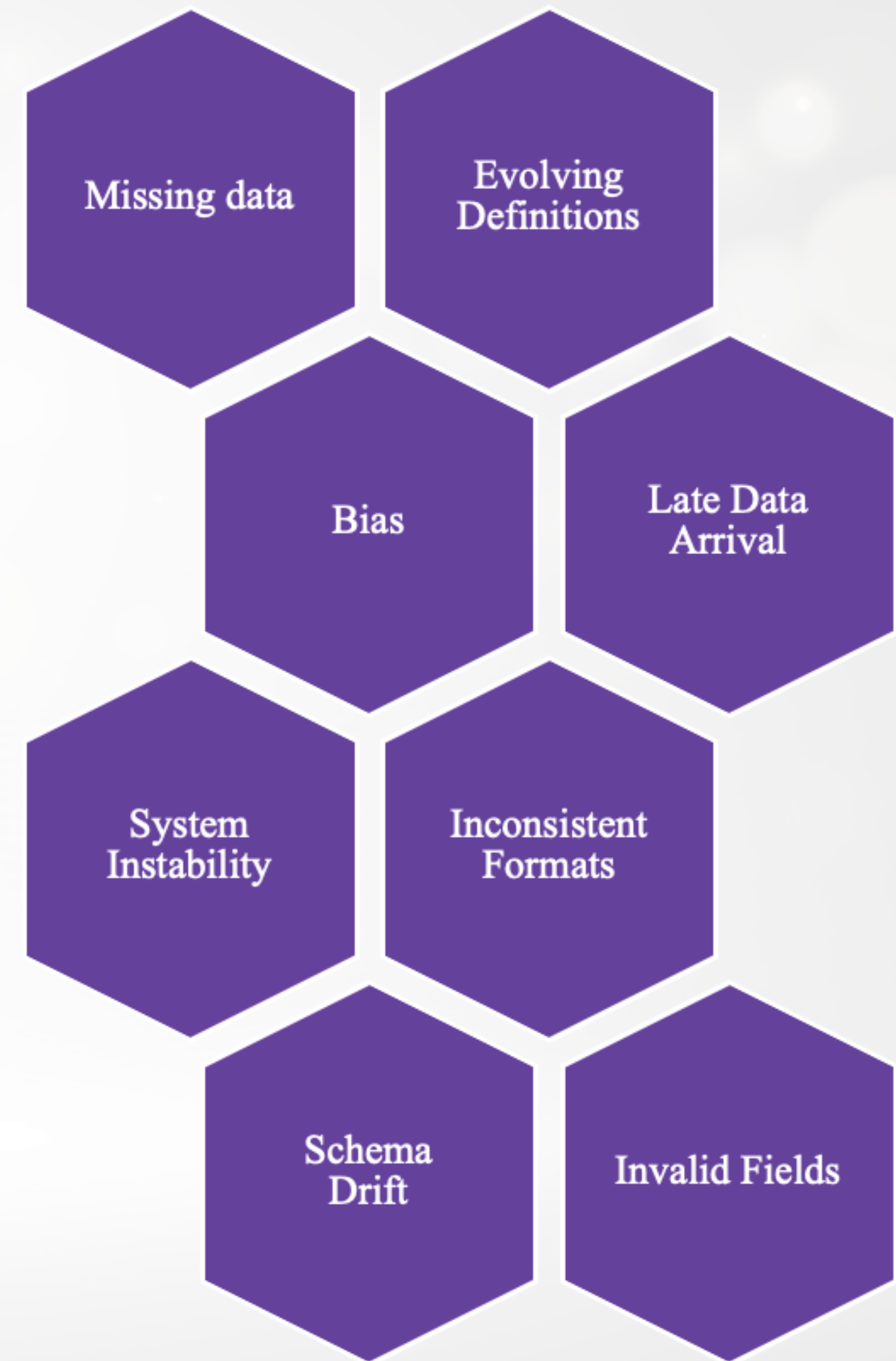
- Waterfall collects feedback early in the project
- Analytics feedback is meaningful only after results exist
- Stakeholders cannot predict needs without outputs
- Late feedback leads to rework
- Communication becomes inefficient



ARGUMENT 03

UNCERTAIN DATA

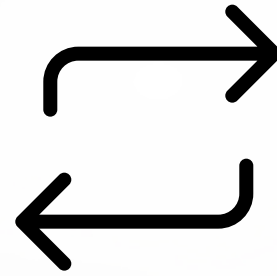
- Waterfall assumes stable and known inputs
- Analytics data issues appear during exploration
- Common issues:
- Missing values
- Inconsistent formats
- Bias and outdated records
- Data uncertainty forces redesign under Waterfall



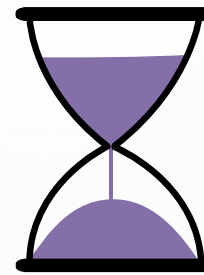
WHY WATERFALL FAILS FOR ANALYTICS?



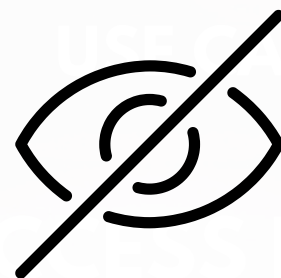
Analytics is learning driven



Change is normal, not exceptional



Feedback comes late by nature



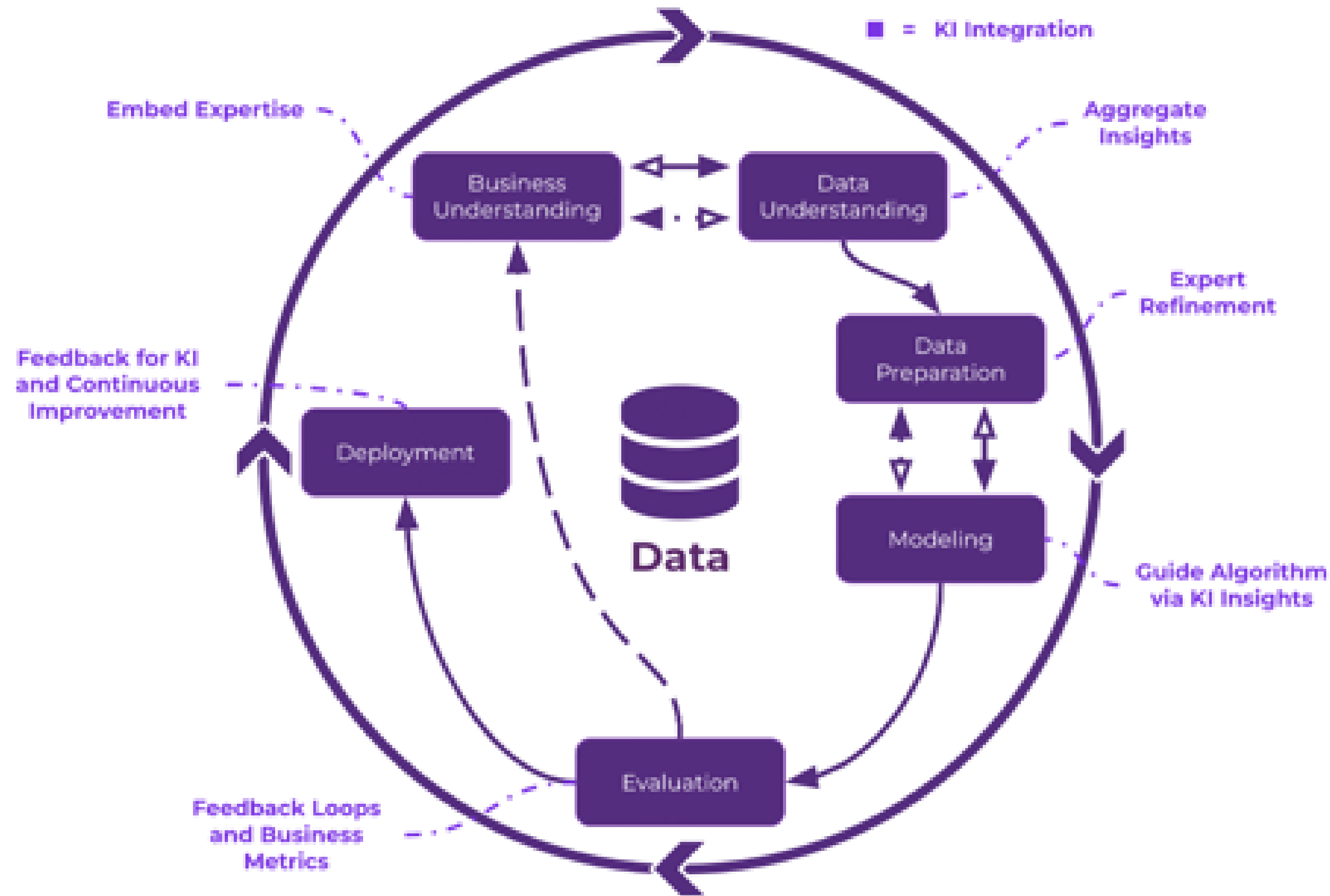
Data cannot be fully known upfront



Waterfall assumptions do not hold

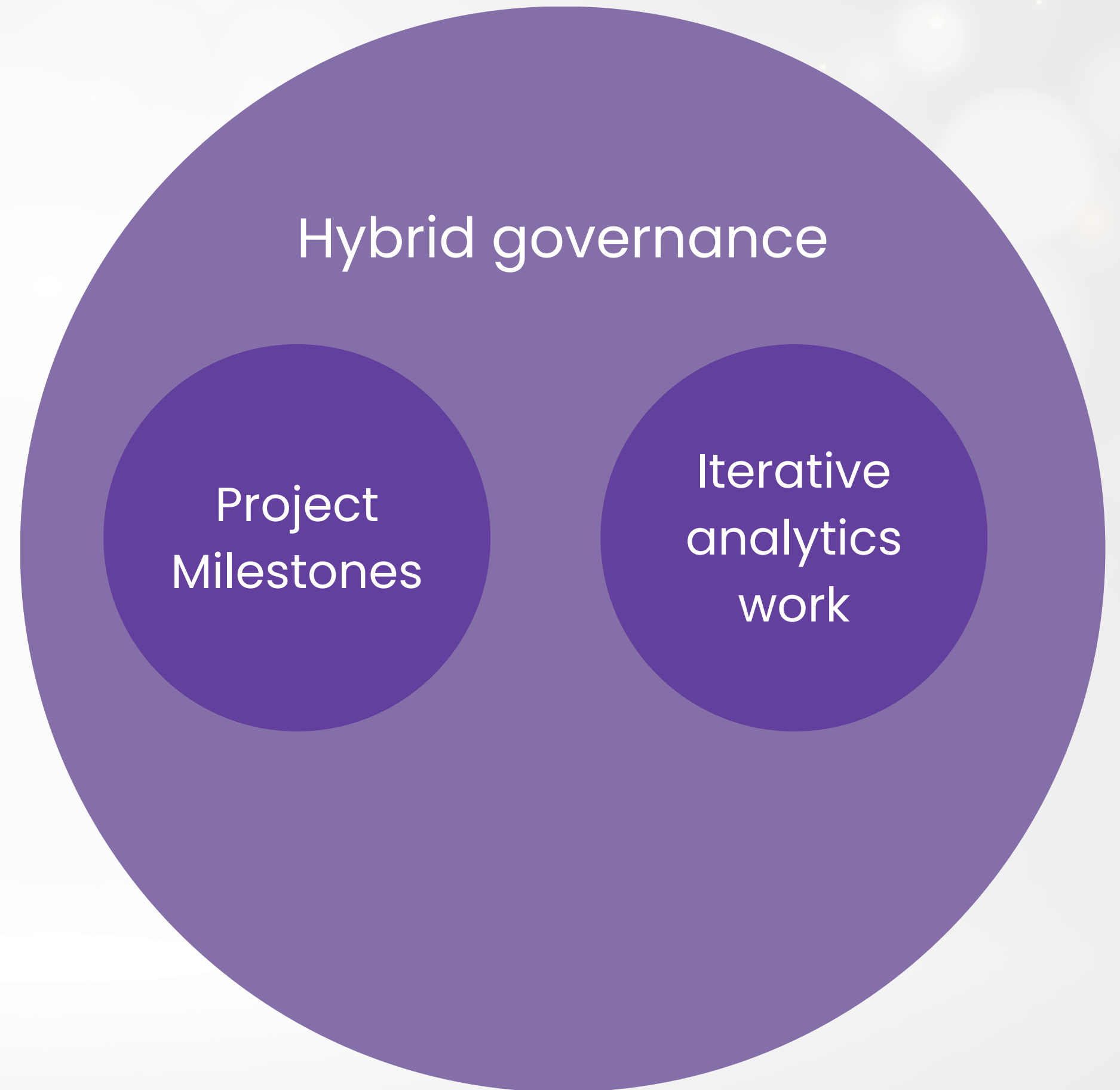
IMPLICATIONS CRISP-DM

- CRISP-DM is iterative and cyclical
- Supports evolving requirements
- Encourages feedback after results
- Handles data uncertainty explicitly
- Aligns with how analytics creates value



IMPLICATIONS HYBRID GOVERNANCE

- Organizations still need structure
- Hybrid governance combines:
- Project milestones
- Iterative analytics work
- Maintains control without restricting learning
- Balances flexibility and accountability



KEY TAKEAWAYS

Waterfall works for stable problems

- Effective when requirements are known in advance
- Assumes limited change during execution
- Best suited for engineering and infrastructure projects

Methodology must match the problem type

- Using the wrong methodology creates friction
- Rigid planning limits learning in analytics
- Project success depends on methodological fit

Analytics projects are uncertain and evolving

- Insights emerge only after data exploration
- Stakeholders refine goals after seeing results
- Change is a normal part of analytics work

Iterative approaches improve outcomes

- Support learning and refinement over time
- Enable meaningful stakeholder feedback
- Reduce rework and misalignment

CONCLUSION

- Waterfall project management is effective when requirements, feedback, and inputs are stable
- Analytics projects operate under fundamentally different conditions
- Requirements evolve through exploration, feedback becomes meaningful only after results, and data uncertainty is unavoidable
- Because of this mismatch, Waterfall struggles when applied to analytics work
- Iterative methodologies such as CRISP-DM better support learning and adaptation
- Hybrid governance models provide a practical way to balance flexibility with organizational control





THANK YOU!

